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Management system of gaming chips and storage box

Abstract

A system that managing playing cards to which information is given by an RFID tag and gaming chips to which information is given by an RFID tag. A system is provided with a storage box integrally configured to contain a card drawer for storing playing cards and a chip drawer for storing gaming chips, a card RFID reader for reading information of an RFID tag attached to the playing cards stored in the card drawer, and a chip RFID reader for reading information of the RFID tag attached storing gaming chips stored in the chip drawer. A card RFID reader and a chip RFID reader reads information of each RFID tag using different frequencies.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5651548	12/1996	French	N/A	N/A
5735742	12/1997	French	N/A	N/A
5831527	12/1997	Jones, II et al.	N/A	N/A
5957776	12/1998	Hoehne	N/A	N/A
7258612	12/2006	Fujimoto	N/A	N/A
7846020	12/2009	Walker et al.	N/A	N/A
8088010	12/2011	Hill	N/A	N/A
8235825	12/2011	French	N/A	N/A
8668148	12/2013	Gronau	N/A	N/A
8915782	12/2013	Fujimoto	463/43	G07F 17/3232
9242436	12/2015	Hallman et al.	N/A	N/A
9858756	12/2017	Gelinotte et al.	N/A	N/A
10493357	12/2018	Shigeta	N/A	N/A
10675530	12/2019	Shigeta	N/A	N/A
11126908	12/2020	Stewart	N/A	G06K 7/10316
2003/0102970	12/2002	Creel et al.	N/A	N/A
2004/0137977	12/2003	Fujimoto	N/A	N/A
2004/0207156	12/2003	Soltys	273/292	G07F 17/3293
2005/0054408	12/2004	Steil	463/11	A63F 1/18
2006/0205484	12/2005	Nicastro	N/A	N/A
2006/0258442	12/2005	Ryan	N/A	N/A
2007/0023500	12/2006	DeRaedt et al.	N/A	N/A

2007/0024449	12/2006	Bilyeu et al.	N/A	N/A
2007/0026949	12/2006	Charlier et al.	N/A	N/A
2007/0052167	12/2006	Galatan	N/A	N/A
2007/0060311	12/2006	Rowe et al.	N/A	N/A
2007/0225055	12/2006	Weisman	273/149P	A63F 1/14
2008/0180250	12/2007	Steil	N/A	N/A
2008/0224400	12/2007	Gnat	N/A	N/A
2009/0048026	12/2008	French	N/A	N/A
2010/0093428	12/2009	Mattice et al.	N/A	N/A
2010/0093429	12/2009	Mattice et al.	N/A	N/A
2010/0304841	12/2009	Sammon et al.	N/A	N/A
2011/0300917	12/2010	Hill	N/A	N/A
2012/0050130	12/2011	Nishioka et al.	N/A	N/A
2012/0080845	12/2011	Emori et al.	N/A	N/A
2012/0100901	12/2011	Kirsch	N/A	N/A
2012/0105215	12/2011	Gronau et al.	N/A	N/A
2012/0241344	12/2011	Gronau et al.	N/A	N/A
2013/0196777	12/2012	Hill	N/A	N/A
2013/0207345	12/2012	Gelinotte	273/309	G07F 17/3202
2013/0316797	12/2012	Gelinotte et al.	N/A	N/A
2014/0033660	12/2013	Shigeta	N/A	N/A
2014/0274344	12/2013	Kelly et al.	N/A	N/A
2014/0291399	12/2013	Koyama	N/A	N/A
2014/0339107	12/2013	Gelinotte et al.	N/A	N/A
2015/0042042	12/2014	Shigeta	N/A	N/A
2015/0087371	12/2014	Hill	N/A	N/A
2016/0180633	12/2015	Yamaguchi et al.	N/A	N/A
2016/0284160	12/2015	Oh	N/A	N/A
2017/0018138	12/2016	Mcburnie	N/A	N/A
2017/0032240	12/2016	Stewart	N/A	G06K 7/10316
2017/0061735	12/2016	Kotab	N/A	N/A
2017/0106270	12/2016	Shigeta	N/A	N/A
2018/0075690	12/2017	Moore et al.	N/A	N/A
2018/0185741	12/2017	Shigeta	N/A	N/A
2018/0189921	12/2017	Feng et al.	N/A	N/A
2018/0190064	12/2017	Feng et al.	N/A	N/A
2018/0253937	12/2017	Curelea	N/A	N/A
2019/0130242	12/2018	Fu	N/A	N/A
2020/0122018	12/2019	Shigeta	N/A	N/A
2021/0335082	12/2020	Shigeta	N/A	N/A
2023/0245526	12/2022	Shigeta	N/A	N/A
2023/0338820	12/2022	Shigeta	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2005205742	12/2006	AU	N/A
2014210600	12/2014	AU	N/A
1508744	12/2003	CN	N/A

102441273	12/2011	CN	N/A
104368144	12/2014	CN	N/A
105473466	12/2015	CN	N/A
2545967	12/2012	EP	N/A
2002-181672	12/2001	JP	N/A
2005-342175	12/2004	JP	N/A
2007222529	12/2006	JP	N/A
2008013332	12/2007	JP	N/A
2009086984	12/2008	JP	N/A
4647247	12/2010	JP	N/A
2012-075781	12/2011	JP	N/A
2012075181	12/2011	JP	N/A
2012-200309	12/2011	JP	N/A
2013-544005	12/2012	JP	N/A
2014-504164	12/2013	JP	N/A
2014-203139	12/2013	JP	N/A
2015-037550	12/2014	JP	N/A
2015139611	12/2014	JP	N/A
2015-525391	12/2014	JP	N/A
2016-116544	12/2015	JP	N/A
2023112690	12/2022	JP	N/A
1020070009132	12/2013	KR	N/A
101468139	12/2013	KR	N/A
2003082420	12/2002	WO	N/A
2007062892	12/2006	WO	N/A
2012161422	12/2011	WO	N/A
2015145499	12/2014	WO	N/A
2016014936	12/2015	WO	N/A
2016187164	12/2015	WO	N/A
2018123929	12/2017	WO	N/A

OTHER PUBLICATIONS

International Search Report issued on Nov. 7, 2017 in corresponding PCT Application PCT/JP2017/032553. cited by applicant

Macao Office Action issued on Jul. 9, 2019 in corresponding MO Application I/001542. cited by applicant

Jieun Yu and Wonjun Lee, "Gentle: Reducing Reader Collision in Mobile RFID Networks", The 4th International Conference on Mobile Ad-hoc and Sensor Networks, MNS 2008, ISBN: 978-0-7695-3457-2, No. XP 031376554, pp. 280-287, Dec. 10, 2008, IEEE Computer Society, Piscataway, NJ, USA. cited by applicant

Australian application number 2017389188, Examination Report No. 1 issued on May 5, 2020. cited by applicant

European application No. 17888889.7, Search Report issued on May 26, 2020. cited by applicant

European application No. 17887313.9, Search Report issued on May 26, 2020. cited by applicant

Japanese application No. 2018-558809, Office Action issued on Jun. 2, 2020. cited by applicant

U.S. Appl. No. 16/474,856, Office Action issued on Jun. 22, 2020. cited by applicant

CN application No. 201711423383.3, Office Action issued on Oct. 15, 2020. cited by applicant

KR application No. 10-2019-0108866, Office Action issued on Dec. 16, 2020 above. cited by applicant

CA application No. 2989878, Office Action issued on Jan. 27, 2021. cited by applicant

U.S. Appl. No. 16/474,856, Office Action issued on Feb. 24, 2021. cited by applicant
U.S. Appl. No. 16/474,856, Office Action issued on Aug. 27, 2021. cited by applicant
JP Office Action issued for JP2018-559435 on Dec. 14, 2021. cited by applicant
US Office Action issued for U.S. Appl. No. 16/474,856 on Apr. 18, 2022. cited by applicant
Korean Office Action issued for KR10-2022-0062341 on Jul. 4, 2022. cited by applicant
Chinese Office Action issued for CN201780086822.8 on Sep. 14, 2022. cited by applicant
US Office Action issued for U.S. Appl. No. 16/474,856 on Oct. 28, 2022. cited by applicant
Chinese Office Action issued for CN 201780087769.3 on Jan. 28, 2023. cited by applicant
Chinese Office Action issued for CN 201780086822.8 on Jun. 10, 2023. cited by applicant
Report on Advances in Electronics and Information Technology 2008-2009 by China Association
for Science and Technology, p. 124-126, published on Mar. 31, 2009. cited by applicant
Japanese Office Action issued for JP2022-156703 on Jul. 4, 2023. cited by applicant
New Zealand Patent Examination Report 1 dated Jan. 25, 2024 issued in NZ Application No.
754985. cited by applicant
U.S Office Action dated Mar. 13, 2024 issued in U.S. Appl. No. 17/335,386. cited by applicant
Australian Examination Report dated Feb. 19, 2025 issued in AU Application No. 2023206173.
cited by applicant
Japanese Office Action dated Mar. 18, 2025 issued in JP Application No. 2023-179750. cited by
applicant
Korean Office Action dated Apr. 23, 2025 issued in KR Application No. 10-2024-0065322. cited by
applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/703,251 filed Mar. 24, 2022, which is a continuation of U.S. patent application Ser. No. 16/474,809 filed Jun. 28, 2019 (now U.S. Pat. No. 11,344,790), which is a national phase application under 35 U.S.C. § 371 of International Application No. PCT/JP2017/032553 filed Sep. 8, 2017, which claims the benefit of JP Pat. App. No. 2016-258007 filed Dec. 30, 2016, the entire contents of each disclosure are incorporated herein by reference.

FIELD

(1) The present invention relates to a system that manages playing cards and gaming chips and a storage box that stores playing cards and gaming chips.

BACKGROUND

(2) Among many table games carried out in casinos and game facilities, baccarat and blackjack are known. In these games, a standard deck composed of 52 playing cards is used and normally distributed from a chute including a plurality of shuffled decks (six to nine or 10 decks) before a game is started. Also in casinos and game facilities, a large number of gaming chips to be used in these games are used.

(3) Gaming chips are put into a chip tray on a game table and after each game ends, the casino side collects and stores won chips (loser's chips) in the chip tray and also pays out the gaming chips to the winner from the chip tray as a settlement of bets.

(4) When gaming chips of the chip tray run short, lacking chips are directly transported from a cage (cashier) of the casino to the chip tray, which is filled therewith. When gaming chips are

transported from the cage to the chip tray, gaming chips are housed in a dedicated case (for example, a case capable of housing 100 chips) for transportation. Conversely, when gaming chips in the chip tray become excessive, excessive chips are directly transported to the cage of the casino after being housed in the dedicated case.

(5) To measure gaming chips inside a casino, a conventional technology that measures the quantity and the amount of money of gaming chips for monitoring by embedding RFID in a gaming chip and reading RFID is known. A technology that manages gaming chips on the game table, gaming chips stored in the chip tray, and gaming chips of the cage by using an RFID attached chip is publicly known and disclosed in Patent Document 1 (U.S. Pat. No. 5,735,742).

(6) It is difficult to prevent fraud and losses only by managing gaming chips in a portion of casinos such as the table, the chip tray, or the cage. Particularly when gaming chips are moved between the chip tray and the cage, it is difficult to prevent theft and losses of gaming chips and so continuous monitoring of gaming chips inside casinos is demanded.

SUMMARY

(7) A system according to an aspect of the present invention is a system for managing a package of shuffled playing cards and gaming chips, wherein the system includes shuffled playing cards including playing cards that constitute a predetermined number of decks and are shuffled in a random order, configured respectively as one container or one package, and having a unique card ID given to the container or the package, a gaming chip provided with a chip ID, a game table on which a game is played by using the shuffled playing cards and the gaming chips, a card storage box storing a plurality of the shuffled playing cards, the shuffled playing cards being conveyed from a card room and configured to be attached to a card shooter apparatus on the game table, a chip storage box storing a plurality of the gaming chips used on the game table and being provided with an opening/closing lock mechanism, and the card ID and the chip ID are identified using wireless tags, and the card storage box and the chip storage box are provided with one or more card readers for reading the card IDs of all the stored shuffled playing cards and one or more chip readers for reading the chip IDs of all the stored chips, and the card readers and the chip readers are respectively configured of readers that use frequencies or modes that differ from each other, by which reading error of the card ID and the chip ID by the wireless tags is minimized.

(8) In the above system, the card ID uses a UHF wireless tag and the chip ID uses a HF wireless tag.

(9) In the above system, further comprising control unit connected to the card reader and the chip reader and configured to detect a change when information read by the card reader or information read by the chip reader is changed.

(10) In the above system, the card reader read information of the wireless tag by a radio wave method, and the chip reader read information of the wireless tag by an electromagnetic induction method.

(11) Another aspect of the present invention is a system for managing playing cards to which card information is given by a card RFID tag and gaming chips to which chip information is given by a chip RFID tag, including storage unit in which card storage unit for storing the playing cards and chip storage unit for storing the gaming chips are included and integrally configured, a card RFID reader that reads information of a RFID tag given to the playing cards stored in the card storage unit, a chip RFID reader that reads information of a RFID tag given to the gaming chips stored in the chip storage unit, wherein the card RFID reader and the chip RFID reader read the information of the respective RFIDs by using different frequencies.

(12) In the above system, the card information is given to a package constituted of a plurality of the playing cards, and the chip information is given to each of the gaming chips.

(13) In the above system, the card RFID tag is attached to the package, and the chip RFID tag is embedded in the gaming chips.

(14) In the above system, the card RFID tag is embedded in the playing cards, and the card

information is given to each of the playing cards.

(15) In the above system, comprising a control unit connected to the card RFID reader and the chip RFID reader and configured to detect a change when information read by the card RFID reader or information read by the chip RFID reader is changed.

(16) In the above system, the card RFID reader reads information of the card RFID tag by a radio wave method, and the chip RFID reader reads information of the chip RFID tag by an electromagnetic induction method.

(17) Still another aspect of the present invention is a storage box for storing playing cards to which card information is given by a card RFID tag and gaming chips to which chip information is given by a chip RFID tag, the storage box comprising card storage unit for storing the playing cards, chip storage unit for storing the gaming chips, a card antenna for reading the card information of the playing cards stored in the card storage unit; and a chip antenna for reading the chip information of the gaming chips stored in the chip storage unit, wherein the card antenna and the chip antenna read information using radio waves with frequencies that differ from each other.

(18) The foregoing and other objects, features, aspects and advantages of the exemplary embodiments will become more apparent from the following detailed description of the exemplary embodiments when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram providing an overview of a table game system according to a first embodiment of the present invention;

(2) FIG. 2A is a perspective view showing a chip case according to an embodiment of the present invention;

(3) FIG. 2B is a perspective view showing a chip case according to a modification of the first embodiment of the present invention;

(4) FIG. 3 is a diagram providing an overview of an overall system that manages the chip case housing gaming chips according to the first embodiment of the present invention;

(5) FIG. 4 is a perspective view showing the gaming chips according to the first embodiment of the present invention;

(6) FIG. 5 is a front sectional view showing the gaming chip according to the first embodiment of the present invention;

(7) FIG. 6 is a perspective view of a package used for a card distribution chute of a management system according to the first embodiment of the present invention and unpacked shuffled playing cards;

(8) FIG. 7 is a perspective view showing a plurality of packages according to a second embodiment of the present invention and a storage box storing a plurality of chip cases according to an embodiment of the present invention;

(9) FIG. 8 is a diagram providing an overview of a system that manages packages of shuffled playing cards and chip cases housing gaming chips according to the second embodiment of the present invention.

(10) FIG. 9 is a diagram providing an overview of the system that manages packages of shuffled playing cards and chip cases housing gaming chips according to the second embodiment of the present invention.

(11) FIG. 10 is a diagram showing the configuration of a system according to a third embodiment of the present invention;

(12) FIG. 11 is a perspective view of an example of a drawer for cards according to the third embodiment of the present invention;

(13) FIG. 12 is a plan view of another example of the drawer for cards according to the third embodiment of the present invention;

(14) FIG. 13 is a plan view of an example of a drawer for gaming chips according to the third embodiment of the present invention; and

(15) FIG. 14 is an explanatory diagram showing movement of cards 6 in a management system using the storage box according to the third embodiment of the present invention.

DETAILED DESCRIPTION OF NON-LIMITING EXAMPLE EMBODIMENTS

(16) A first embodiment provides a system that monitors gaming chips of a casino in real time and the system can handle an error when the error occurs by managing all gaming chips in the casino. The first embodiment further provides a system of continuous monitoring that, in addition to management of gaming chips, prevents losses because if a shuffled playing card package is once lost in a casino, someone may know the alignment thereof and the package may not be used in a game.

(17) A system that manages gaming chips according to the first embodiment of the present invention will be described below. FIG. 1 is an explanatory diagram providing an overview of a table game of casino according to an embodiment of the present invention. In the present embodiment, a game table 21 includes a betting area 24 where a betting person 2 places a gaming chip 3 as a bet and a chip tray 17 capable of housing a plurality of gaming chips to collect a lost chip 3L and redeem a won chip 3W after each game ends. Also, a card shooter apparatus 25 placed on the game table 21, having a card reader that reads the number (rank) of a mark of the card 6, and having a controller 27 that determines the winner according to rules of a table game based on information of the number (rank) of the card 6 successively read by the card shooter apparatus 25 is installed.

(18) An increase/decrease amount of the gaming chips 3 in the chip tray 17 before/after collection of the lost chip 3L and redemption of the won chip 3W can be calculated by comparing the total of the gaming chips 3 in the chip tray 17 before collection of the lost chip 3L and redemption of the won chip 3W and the total of the gaming chips 3 in the chip tray 17 after collection of the lost chip 3L and redemption of the won chip 3W. The total of the gaming chips 3 in the chip tray 17 before collection of the lost chip 3L and redemption of the won chip 3W and the total of the gaming chips 3 in the chip tray 17 after collection of the lost chip 3L and redemption of the won chip 3W can be detected by embedding RFID indicating its quantity in the gaming chip 3 and providing an RFID reader 18 in the chip tray 17.

(19) FIG. 2A is a perspective view of a dedicated chip case 100 to house a plurality of gaming chips 3 used in casinos. The chip case 100 includes an upper portion 101 and a lower portion 102. In the present embodiment, the upper portion 101 and the lower portion 102 are made of transparent resin. A light transmission portion that allows light to transmit may also be provided so as to be able to image housed gaming chips using a camera. Also in the present embodiment, the case 100 has a sealing structure made of the upper portion 101 and the lower portion 102, but the light transmission portion may be in a perforated state.

(20) In the present embodiment, the case 100 has a shape in which five columns, in each of which 20 pieces of the gaming chip 3 are overlaid and housed, are formed in parallel and in the example of FIG. 2A, the cross section of each column is polygonal (octagonal) so that the upper portion 101 and the lower portion 102 roughly match the shape of a gaming chip C.

(21) A unique chip case ID code 103 is attached to the chip case 100. The chip case ID code 103 is related to a chip ID code 4 of the gaming chip 3 housed in the chip case.

(22) FIG. 2B is a perspective view of a case 100' according to a modification. The case 100' is also made of an upper portion 101' and the lower portion 102 constructed of transparent resin. In the present modification, the top surface is formed from a flat surface. By changing to the flat surface, a linear shadow due to edges of a polygonal cross section of the case 100' does not appear in the camera so that information of the side face of the gaming chip 3 can correctly be identified in

image analysis of a shot image of the camera.

(23) FIG. 3 is an explanatory diagram providing an overview of a management system of gaming chips installed beside the game table 21 and using a storage box 200 storing a plurality of the chip cases 100 in which the gaming chips 3 are housed. The gaming chips 3 for replenishment are housed in the storage box 200 while being put into the chip case 100 and when the gaming chips 3 in the chip tray 17 run short, a dealer 11 takes out the chip case 100 together with the gaming chips 3 for replenishment from the storage box 200 to set the chip case 100 to the chip tray 17. When the gaming chips 3 in the chip tray 17 become excessive, the excessive gaming chips 3 can be put into the chip case 100 and housed in the storage box 200. Thus, the storage box 200 is placed by the dealer 11 beside the game table 21.

(24) When the chip cases 100 stored in the storage box 200 and housing the gaming chips 3 run short, as many the chip cases 100 as necessary can be moved from a cage (cashier) 201 in a casino for replenishment. When the chip cases 100 stored in the storage box 200 become excessive, the chip cases 100 that are excessive are moved to the cage 201.

(25) The chip case ID code 103 is attached to the chip case 100 and the chip case ID code 103 attached to the chip case 100 is continuously read by one or a plurality of readers for reading chip case ID 202 installed inside the storage box 200.

(26) The storage box 200 also includes one or a plurality of chip readers that reads the chip ID code 4 of all the gaming chips 3 stored.

(27) A control apparatus 204 has a function to output the total number of the gaming chips 3 stored in the storage box and all the chip ID codes 4 stored in the storage box by monitoring the chip ID code 4 read by a chip reader 203.

(28) The control apparatus 204 has a function to output whether an increase/decrease amount or increase/decrease value of the gaming chips 3 placed in the chip tray 17 of the game table 21 or an increase/decrease amount or increase/decrease value of the gaming chips 3 placed in the cage 201 that manages the gaming chips 3 of a casino and an increase/decrease amount or increase/decrease value of the gaming chips 3 housed in the storage box 200 match.

(29) Whether the chip case 100 placed in the storage box 200 is inside the storage box 200 may be monitored by the control apparatus 204 at fixed intervals (for example, every one minute, every five minutes, every one hour or more). The storage box 200 may have the reader for reading chip case ID 202 to read the chip case ID code 103 of the chip case 100 arranged in a drawer 205 of the storage box 200. The reader for reading chip case ID 202 and the chip reader 203 of the storage box 200 may be a bar code reader R (alternatively, an RFID tag reader or QR code (registered trademark) reader (not shown) may be used instead of the bar code reader R). The reader for reading chip case ID 202 and the chip reader 203 may be installed so as to be able to scan in the X direction and Y direction to read all ID codes of the chip cases 100 by a scan unit 53 installed in the drawer 205. Also, a transmission unit 206 to transmit information obtained by the reader for reading chip case ID 202 and the chip reader 203 to the outside of the storage box 200 is provided. The storage box 200 has a lock unit 207 to prevent the chip case 100 from being taken out from the storage box 200 by opening the drawer 205. The lock unit 207 is unlocked only while an authorized person of a casino puts in or takes out the chip case 100 from the storage box 200. Only an authorized person of a casino can operate the lock unit 207.

(30) The storage box 200 includes the lock unit 207 to prevent the drawer 205 from opening and the lock unit 207 may include a warning unit (may be wireless) to notify that the drawer 205 has opened. When a notification that the drawer 205 has opened is received (or when appropriate), the storage box 200 may be imaged by a nearest monitoring camera 29 to record taking in or out of the chip case 100 from the storage box 200 by an authorized person or others. By recording such behavior (images in which the storage box 200 is opened), the fact that the chip case 100 can be taken in or out from the storage box 200 only while the lock unit 207 is unlocked by an authorized person (while the drawer 205 is opened) can be confirmed. By monitoring such images, the

presence of all the chip cases **100** inside the storage box **200** can be confirmed.

(31) The storage box **200** may have the plurality of readers **202** to read the chip case ID code **103** in an upper portion inside the storage box **200**. The storage box **200** has the drawer **205** and the chip case **100** can be taken in or out by opening or closing the drawer **205**. The storage box **200** includes the lock unit **207** to prevent illegal pilfering of the chip case **100**. Further, as another idea, the storage box **200** may include as many the readers **202** as the maximum number of the chip cases **100** that can be stored in an upper portion inside the storage box **200**.

(32) Next, a gaming chip according to an embodiment of the present invention will be described. FIG. **4** is a perspective view of a state in which the gaming chips **3** are stacked and FIG. **5** is a sectional side elevation of the gaming chip **3**. As shown in FIG. **4**, the gaming chip **3** has at least a 5-layer structure in which a common color layer **122** has printing **123** (such as 100 points) indicating the kind (value) of the gaming chip **3** done on the surface (the upper surface and the undersurface), a transparent layer **120** is provided on the outermost layer, and layers are thermocompression-bonded. These gaming chips **3** are formed by using a plastic material in a long and narrow shape, forming a closely adhering condition (such as a 5-layer structure) in which each layer (a designated color layer **121**, the common color layer **122**, and the transparent layer **120**) is thermocompression-bonded in a long state, and then stamping into a circular shape or rectangular shape by press or the like. An R finish (round angle) is given to edges of the transparent layer **120** in the outermost layer by designing dimensions of the die and punch of the mold to stamp when stamped by press.

(33) Further, the gaming chip **3** is provided with a mark M in UV ink or carbon black ink on the surface of the common color layer **122**. The mark M indicates genuineness of the gaming chip **3** and becomes visible when an ultraviolet ray (or an infrared ray) is applied thereto to indicate authenticity by its shape or a combination of numbers. The transparent layer **120** is thermocompression-bonded or coated as the outermost layer like covering the printing **123** and the mark M and the transparent layer **120** are embossed to prevent the gaming chips **3** from coming into close contact with each other.

(34) An R finish (R) is given to edges of the transparent layer **120** in the outermost layer where the printing **123** (such as 100 points) is done to prevent the surface of the common color layer **122** from appearing on the side face after being discolored during the stamping process of gaming chip **3**. Also, the R finish prevents hands and other gaming chips **3** from being damaged by otherwise remaining sharp edges of the gaming chip **3**.

(35) The designated color layer **121** may be formed from, as shown in FIG. **5**, a plurality (three layers in FIG. **5**) of layers colored in the designated color. The plurality (three layers in FIG. **5**) of layers colored in the designated color is thermocompression-bonded to each other and thus, the 3-layer structure is not visible as shown in FIG. **5** and FIG. **5** shows the three layers of the designated color layer **121** from a description viewpoint. Further, a hollow B is partially provided in the center layer of the three layers of the designated color layer **121** and an RFID tag **125** is contained therein.

(36) As shown in FIGS. **4** and **5**, the gaming chip **3** has a stacked multilayer structure and a striped pattern in a lamination direction is clearly formed on the side face so that, when compared with a conventional gaming chip, the color (the kind of gaming chip) of the designated color layer **121** and the number can be measured easily and correctly by the image analysis. The side face of the gaming chip **3** can be photographed by a camera so that the designated color layer **121** can clearly be identified. Further, if, in addition to the image analysis, an AI-utilizing computer or control system and deep learning (structure) technology are used, the analysis and determination of images can be made more correct. The AI-utilizing computer or control system and deep learning (structure) technology are already known and available to persons skilled in the art and so a detailed description thereof is omitted.

(37) Next, an overview of a management system of gaming chips and shuffled playing cards using the storage box according to a second embodiment of the present invention will be described. In the

second embodiment of the present invention, both of gaming chips and packages of shuffled playing cards are managed.

(38) FIG. 6 is an explanatory diagram providing an overview of a package of shuffled playing cards used in a table game of a casino and aligned randomly being used in the casino according to an embodiment of the present invention. In the present embodiment, shuffled playing cards **301S** are packed as a package **302** and the package **302** is unpacked so that the shuffled playing cards **301S** can be used for games on the table and set to the card shooter apparatus. During the game, the dealer **11** draws out cards **301** from the card shooter apparatus **25** and distributes the cards **301** to the game table **21**. Cards of each shuffled playing card **301S** with a predetermined number of decks (usually 6, 8, 9 or 10 decks) are produced so as to be randomly shuffled and uniquely randomly aligned individually and packed together with a package ID code **304** attached to the package **302** as a bar code **303** individually identifiable by the bar code reader R. RFID (or an RF tag) may be attached to the ID code **304** instead of or together with the bar code **303**.

(39) FIGS. 7 and 8 are explanatory diagrams providing an overview of a management system of gaming chips and packages of shuffled playing cards using the storage box according to the second embodiment of the present invention. The embodiment of the present invention provides a management system of the gaming chips **3** and the packages **302** of the shuffled playing cards **301S** to play a game (baccarat). The gaming chips **3** for replenishment are housed in the storage box **200** while being put into the chip case **100** and when the gaming chips **3** in the chip tray **17** run short, the dealer **11** takes out the chip case **100** together with the gaming chips **3** for replenishment from the storage box **200** to set the chip case **100** to the chip tray **17**. When the gaming chips **3** in the chip tray **17** become excessive, the excessive gaming chips **3** can be put into the chip case **100** and housed in the storage box **200**. The storage box **200** also houses the package **302** to be used for the next game and the dealer **11** takes out the package **302** to be used for the next game from the storage box **200** and sets the shuffled playing cards **301S** to the card shooter apparatus **25**.

(40) The chip case ID code **103** is attached to the chip case **100** and the chip case ID code **103** attached to the chip case **100** is continuously read by the reader for reading chip case ID **202** installed inside the storage box **200**. Also, the package ID code **304** is attached to the package **302** and the package ID code **304** attached to the package **302** is continuously read by a reader for reading package ID **305** installed inside the storage box **200**.

(41) The storage box **200** includes one or a plurality of the readers for reading package ID **305** that reads the playing card ID code of all stored shuffled playing cards and also one or a plurality of the chip readers **202**, **203** that reads the case ID code of all the stored chip cases.

(42) The control apparatus **204** has a function to output the total numbers of the shuffled playing cards **301S** and the chip cases **100** stored in the storage box **200** and all the package ID codes **304** and all the chip case ID codes **103** stored in the storage box **200** by monitoring the playing card ID code read by the reader for reading package ID **305** and the chip case ID code **103** read by the chip reader **203**.

(43) The control apparatus **204** has a function to grasp an increase/decrease of the chip cases **100** housed in the storage box **200** by periodically monitoring the case ID code **103** of the chip case **100** and, when the increase/decrease is grasped, to output the total amount after the increase/decrease of value of all the gaming chips **3** housed in the storage box **200**.

(44) Whether the chip case **100** and the package **302** placed in the storage box **200** are inside the storage box **200** may be monitored by the control apparatus **204** at fixed intervals (for example, every one minute, every five minutes, every one hour or more). The storage box **200** may have the reader for reading package ID **305** to read the ID code **304** of the package **302** of shuffled playing cards and the reader for reading chip case ID **202** to read the chip case ID code **103** of the chip case **100** arranged in the drawer **205** of the storage box **200**. The reader for reading package ID **305** and the reader for reading chip case ID **202** of the storage box **200** may be a bar code reader R or the monitoring camera **29** (alternatively, an RF ID tag reader or QR code (registered trademark) reader

(not shown) may be used instead of the bar code reader R). The reader for reading package ID **305** and the reader for reading chip case ID **202** may be installed so as to be able to scan in the X direction and Y direction to read all the ID codes **304** of the packages **302** by the scan unit **53** installed in the drawer **205**. Also, a transmission unit **206** to transmit information obtained by the reader for reading package ID **305** and the reader for reading chip case ID **202** to the outside of the storage box **200** is provided. The storage box **200** has the lock unit **207** to prevent the package **302** from being taken out from the storage box **200** by opening the drawer **205**. The lock unit **207** is unlocked only while an authorized person of a casino takes in or out the package **302** from the storage box **200** (the drawer **205** is opened). Only an authorized person of a casino can operate the lock unit **207**.

(45) The storage box **200** includes the lock unit **207** to prevent the drawer **205** from opening and the lock unit **207** may include a warning unit (may be wireless) to notify that the drawer **205** has opened. When a notification that the drawer **205** has opened is received (or when appropriate), the storage box may be imaged by the nearest monitoring camera **29** to record taking in or out of the package **302** of the chip case from the storage box **200** by an authorized person or others. By recording such behavior (images in which the storage box **200** is opened), the fact that the package **302** can be taken in or out from the storage box **200** only while the lock unit **207** is unlocked by an authorized person (while the drawer **205** is opened) can be confirmed. By monitoring such images, the presence of all the packages **302** inside the storage box **200** can be confirmed.

(46) FIG. **9** is a diagram providing an overview of a system that manages packages of shuffled playing cards and chip cases housing gaming chips according to the second embodiment of the present invention. As another embodiment of the storage box **200**, the storage box **200** may have a plurality of the readers for reading package ID **305** to read the package ID code **304** and a plurality of the readers for reading chip case ID **202** to read the chip case ID code **103** in an upper portion inside the storage box **200**. By moving each of the scan unit **53** arranged in an upper portion of the storage box **200** in the Y direction, each ID code reader moves in the Y direction to read all of the package ID code **304** of the package **302** in each column below each ID code reader and the chip case ID code **103** of the chip case **100**. The storage box **200** has a drawer and the package **302** and the chip case **100** can be taken in or out by opening or closing the drawer. The storage box **200** includes a lock unit **56** to prevent illegal pilfering of the package **302** and the chip case **100**. Further, as another idea, the storage box **200** may include as many ID code readers as the maximum numbers of the packages **302** and the chip cases **100** that can be stored in an upper portion inside the storage box **200**.

(47) The present embodiment relates to, like the second embodiment, improvements of technology to read ID of the gaming chip **3** and the playing card **6** in the storage box **200**. Incidentally, matters described in the first or second embodiment can also be applied to the third embodiment.

(48) When RFID tags are attached to different items such as gaming chips and playing cards for management, RFID tags of the same frequency are normally used and content of each item is written into the RFID tag. Accordingly, content of the item can be recognized by reading the RFID tag. In a clothing store, for example, RF tags using radio waves of the same frequency are attached to socks and shirts and information about whether an item is a shirt or socks (information indicating the type of an item) is written into each RF tag. Then, by reading the RF tag, whether the item is socks or a shirt can be recognized.

(49) For security items like gaming chips and playing cards used in casinos, however, the security level demanded may be different from item to item. Also, circumstances in which RFID tags of all kinds of security items can be read by the same reading device (RFID reader) is dangerous and it is desirable to use a separate reading device for each security item.

(50) The present embodiment is developed in view of the above circumstances, and an object thereof is to improve safety when a plurality of types of security items is managed by RFID.

(51) Hereinafter, a system according to the present embodiment will be described specifically with

reference to the drawings. In the description that follows, the description is omitted when appropriate by attaching the same reference signs to the same elements as those in the above embodiments. FIG. 10 is a diagram showing the configuration of a system according to the present embodiment. A system 500 manages the package 302 of the shuffled playing cards 301S and the gaming chips 3. The system 500 includes the game table 21 to play card games and the storage box 200 to store the playing cards 6 and the gaming chips 3 used in card games.

(52) The game table 21 is formed linearly on a side corresponding to the dealer position where the dealer is positioned and formed like an elliptic curve on a side corresponding to the player position where players are positioned. The chip tray 17 to house the gaming chips 3 of the dealer is provided in front of the dealer position of the game table 21. Games are played on the game table 21 using the playing cards 6 and the gaming chips 3.

(53) The chip tray 17 is embedded in the game table 21 by a removable method. The dealer collects the gaming chips 3 bet by the losing player from the game table 21 and houses the gaming chips 3 in the chip tray 17 and then pays out the gaming chips 3 to the winning player from the chip tray 17.

(54) The gaming chip 3 to be used is the same as that described in the first embodiment and contains the RFID tag 125 as a wireless tag and also has a striped pattern on the side face. In the RFID tag 125, chip ID that uniquely identifies the gaming chip 3 and information indicating value of the gaming chip 3 are stored. Also, the color of the striped pattern on the side face indicates value of the gaming chip 3.

(55) The card shooter apparatus 25 is installed on the game table 21. The card shooter apparatus 25 is configured in the same manner as in the first embodiment and playing cards of the predetermined number of decks pulled out from the package 302 are housed in the card shooter apparatus 25 and taken out one by one from an outlet by the dealer to be submitted to a card game.

(56) Playing cards housed in the card shooter apparatus 25 are provided as the package or container (hereinafter, simply called "package") 302. Playing cards constituting the predetermined number of decks are shuffled randomly and individually constituted as the package 302. The package 302 is configured in the same manner as in the first embodiment. In the present embodiment, an RFID tag 306 is attached to the package 302 as a wireless tag. Package ID that uniquely identifies each of the packages 302 is stored in the RFID tag 306. Incidentally, the RFID tag 306 may be embedded or included in the package 302.

(57) The storage box 200 has a cabinet form and has a plurality of drawers. In the present embodiment, an upper drawer is a chip drawer 210 as a chip storage box or chip storage means that houses the gaming chips 3 and a lower drawer is a card drawer 220 as a card storage box or card storage means that houses the packages 302 of shuffled playing cards. That is, the gaming chips 3 and the packages 302 are housed in different drawers of the storage box 200. Also, the storage box 200 is integrally configured by including the chip drawer 210 as a chip storage box and the card drawer 220 as a card storage box.

(58) The chip drawer 210 stores a plurality of the gaming chips 3 used on the game table 21. The card drawer 220 stores a plurality of the packages 302 of shuffled playing cards carried from a card room and inserted into the card shooter apparatus 25 on the game table 21. The chip drawer 210 and the card drawer 220 include the lock unit 207 as an opening/closing lock apparatus.

(59) The storage box 200 is arranged in the dealer position under the game table 21 as a position easy to access from the dealer and difficult to access from players. Also, the storage box 200 is provided with the control unit 204 and the transmission unit 206 similar to those in the first embodiment. The control unit 204 is configured by a management program in the present embodiment being executed by a computer including a storage apparatus. The transmission unit 206 communicates with other devices by wire or by wireless.

(60) FIG. 11 is a perspective view of an example of the card drawer 220. The card drawer 220 of the example in FIG. 11 has a size capable of housing three units of the package 302 in the width

direction and three units in the depth direction, and maximally nine units of the package 302. A card antenna 602 to read the RFID tag 306 attached to the package 302 is contained in the sidewall on the left and right of the card drawer 220. The card antenna 602 may also be affixed to the inner surface of the sidewall.

(61) The storage box 200 further includes a reader for reading package ID 601 connected to the card antenna 602 and the reader for reading package ID 601 is connected to the control unit 204. The reader for reading package ID 601 is an RFID reader and reads information stored in the RFID tag 306 attached to the package 302 via the card antenna 602. The card antenna 602 extends in the depth direction of the sidewall and can read the RFID tag 306 of all the packages 302 housed in the card drawer 220. The card RFID reader 601, the card antenna 602, and the RFID tag 306 attached to the package 302 constitute a card RFID system 600.

(62) FIG. 12 is a plan view of another example of the card RFID system 600. The card drawer 220 of the card RFID system 600 in this example has a size capable of housing three units of the package 302 in the width direction, six units in the depth direction, and maximally 18 units of the package 302. A total of six units of the card antenna 602, two units in the width direction and three units in the depth direction are provided on the undersurface of a member covering the card drawer 220 on the storage box 200 from above (a member partitioning the chip drawer 210 and the card drawer 220). Using the six units of the card antenna 602, the RFID tags 306 attached to 18 units of the package 302 that can be housed.

(63) The six units of the card antenna 602 are connected to the card RFID reader 601 as a reader for reading package ID. In this case, a plurality of the card RFID readers 601 is provided and one of the card RFID readers 601 may be connected to one of the card antennas 602 or one of the card RFID readers 601 may be connected to a plurality of the card antennas 602. Thus, the RFID tags 306 of all the packages 302 housed in the card drawer 220 can be read by one or the plurality of card RFID readers 601.

(64) In the example of FIG. 12, the card antenna 602 is provided in a plate member partitioning the chip drawer 210 and the card drawer 220 in the storage box 200, but instead, the card antenna 602 may be provided at the bottom of the card drawer 220 in the same arrangement as in FIG. 12.

(65) FIG. 13 is a plan view of an example of a chip RFID system 700. In the example of FIG. 13, the chip RFID system 700 has a size capable of housing three units of the chip case 100 in the width direction and two units in the depth direction, and maximally six units of the chip case 100. Incidentally, the chip case 100 is configured in the same manner as the chip case 100 in the first embodiment. That is, the chip case 100 has a plurality of columns housing the gaming chips 3 by being stacked in the thickness direction and the chip case 100 can house 100 pieces of the gaming chips 3.

(66) A pair of left and right chip antennas 702 sandwiching the chip case 100 from both sides are provided in each housing position of the six units of the chip case 100. The chip antenna 702 is provided on the undersurface of a member covering the chip drawer 210 (a top plate of the storage box 200) in the storage box 200 and extends downward. The chip antenna 702 is connected to a chip RFID reader 701 as a reader for reading chip ID. In this case, a plurality of the chip RFID readers 701 is provided and one of the chip RFID readers 701 may be connected to one of the chip antennas 702 or one of the chip RFID readers 701 may be connected to a plurality of the chip antennas 702. Thus, information stored in the RFID tags 125 of all the gaming chips 3 housed in the chip drawer 210 can be read by one or the plurality of chip RFID readers 701. The chip RFID reader 701, the chip antenna 702, and the RFID tag 125 contained in the gaming chip 3 constitute a chip RFID system 700.

(67) In the example of FIG. 13, the chip antenna 702 is provided in the top plate of the storage box 200, but instead, the chip antenna 702 may also be provided on the upper surface of the bottom of the chip drawer 210 in the same arrangement as in FIG. 13.

(68) Hereinafter, control of the control unit 204 will be described. First, the control unit 204 has a

function similar to that in the first and second embodiments. The control unit **204** grasps the number of the packages **302** housed in the card drawer **220** of the storage box **200** and their package IDs based on read results of the RFID tag **306** attached to the package **302** by the card RFID system **600**.

(69) Also, the control unit **204** grasps the number of the gaming chips **3** housed in the chip drawer **210** of the storage box **200**, their chip IDs, and value thereof based on read results of the RFID tag **125** contained in the gaming chip **3** by the chip RFID system **700**. The control unit **204** further determines the total amount of value of the gaming chips **3** housed in the chip drawer **210** based on the number of each value of the gaming chips **3** housed in the chip drawer **210**.

(70) The card RFID system **600** and the chip RFID system **700** periodically read the RFID tag **125** and the RFID tag **306** at predetermined intervals and output read results to the control unit **204** together with table ID that identifies the game table **21**. The control unit **204** monitors read results of the card RFID system **600** and the chip RFID system **700** and, when read results vary, detects the variations and records the table ID and read results in the storage apparatus together with the relevant date and time. Instead, the control unit **204** may record all read results of the card RFID system **600** and the chip RFID system **700** in the storage apparatus together with the relevant date and time and table ID. Alternatively, the control unit **204** and the lock unit **207** may be linked so that the control unit **204** records read results when the lock unit **207** is unlocked.

(71) A warning unit (for example, a warning lamp or an alarm output speaker) may be connected to the control unit **204**. In such a case, all package IDs and chip IDs that can be detected are stored in the storage apparatus of the control unit **204** and the control unit **204** determines whether the read package ID or chip ID matches one of package IDs and chip IDs stored in the storage apparatus. If the read package ID or chip ID matches none of package IDs and chip IDs stored in the storage apparatus, the control unit **204** may control the warning unit to output a warning (for example, a warning lamp is turned on or an alarm is output from an alarm output speaker).

(72) If two units of the package **302** or more decrease at a time (the package **302** is normally fetched one unit at a time) or the number of the gaming chips **3** is not a multiple of 100 (100 pieces of the gaming chips **3**, which is the maximum number that can be housed, are normally housed in the chip case **100** before being housed in the chip drawer **210**), the control unit **204** detects such movement as illegal movement of the package **302** or the gaming chip **3** and may record the movement or output an alarm.

(73) In the storage box **200**, as described above, the housing location of the gaming chips **3** and that of the packages **302** are physically separated, but are only separated inside the storage box **200** and are not apart on the order of meters. Thus, it is necessary to avoid interference between the card RFID system **600** and the chip RFID system **700**.

(74) In the present embodiment, therefore, different frequencies are adopted for the card RFID system **600** and the chip RFID system **700**. A specific example is as follows: In the present embodiment, the electromagnetic induction type is adopted for the chip RFID system **700** and its frequency band used is the HF band (MODE3). The HF band (MODE3) is a short-wave band of 13.56 MHz. The chip antenna **702** is formed in a coil shape. Also, the RFID tag **125** contained in the gaming chip **3** is provided with a coil-shaped antenna. The antenna of the gaming chip **3** transmits/receives radio waves of the HF band to/from the chip antenna **702** and also obtains operating power of the RFID tag **125** by receiving radio waves of the HF band from the chip antenna **702**.

(75) The HF band has a short communication range and directivity and thus, the area to be read can be limited to a predetermined range and reading of the gaming chip **3** in a position that should not be read can intentionally be prevented. When the RFID tag **125** of the gaming chip **3** housed in the chip tray **17** is read, the RFID tag **125** of the gaming chip **3** on the game table **21** can be prevented from being read. Further, when the gaming chip **3** housed in the chip tray **17** is read, each column can be read while avoiding interference between columns by dividing the antenna for each column.

The gaming chips **3** are used and managed by being stacked and by using the HF band, reading can be done even if the gaming chips **3** are stacked and a plurality of the RFID tags **125** are congested. (76) On the other hand, the radio wave type is adopted for the card RFID system **600**. Its frequency band is the UHF band and ultra-high frequencies in the 900 MHz band are used. The card antenna **602** radiates radio waves to space. The RFID tag **306** is also provided with an antenna and radio waves radiated to space are received by this antenna. The UHF band (ultra-high frequency) has higher frequencies than the HF band (short-wave band) and thus, the wavelength becomes shorter, which is advantageous for miniaturization of the antenna. In addition, the UHF band generally has a longer communication range than the HF band.

(77) In the present embodiment, as shown in FIG. **12**, a patch antenna is used as the card antenna **602** of the card RFID system **600** using the UHF band and a dipole antenna is used for the RFID tag **306** of the package **302**. The RFID tag used in the UHF band is generally small and its memory capacity is small and so can be manufactured at low cost. The package **302** is disposed of after playing cards are used together with playing cards and thus, being at low cost is advantageous. Because the communication range of the UHF band is long, even if the packages **302** are put in a carton or further stacked on a pallet, the packages **302** contained in such a carton or pallet can be read together.

(78) FIG. **14** is an explanatory diagram showing movement of the cards **6** in a management system using the storage box according to the third embodiment of the present invention. The playing cards **6** are housed and stored in the package **302** in a card room. The package **302** includes the playing cards **6** (shuffled playing cards **301S**) of eight decks aligned in random order by being shuffled. The storage box **200** is provided for each of the game tables **21**. When the stock of the package **302** in the storage box **200** gets short, the storage box **200** is replenished with the package **302** from the card room. In such a case, the storage box **200** may be replenished with a pallet of a plurality of the packages **302** (for example, nine packages) from the card room.

(79) The card shooter apparatus **25** is installed on the game table **21**. Also, the game table **21** is provided with a disposal port **28** into which the playing card **6** to be disposed of is inserted. In a card game (baccarat in the present embodiment), the playing cards **6** are pulled out one by one from the card shooter apparatus **25** by the dealer and placed on the game table **21**. When one game ends, playing cards **6a** on the game table **21** used for the game are disposed of through the disposal port **28**. When a cut card is drawn from the card shooter apparatus **25**, playing cards **6b** remaining in the card shooter apparatus **25** are disposed of through the disposal port **28**. The playing cards **6** disposed of through the disposal port **28** are transported to the disposal location.

(80) According to the present embodiment, as described above, the card RFID system **600** and the chip RFID system **700** use mutually different frequencies to prevent interference of radio waves with each other and read errors due to interference of package ID attached to the package **302** and chip ID attached to the gaming chip **3** are minimized. To reliably prevent interference, a shielding means (for example, a shielding plate) that blocks radio waves may be provided between chip drawer **210** and the card drawer **220**.

(81) In the above embodiments, card ID that uniquely identifies the package **302** of shuffled playing cards is attached to the package **302**, but as a modification, in place thereof or in addition thereto, the card RFID system **600** may be constructed by causing each of the playing cards **6** to contain the RFID tag. In such a case, card ID that uniquely identifies each of the playing cards **6** is stored in the RFID tag contained in each of the playing cards **6**.

(82) In the above embodiments, chip ID that uniquely identifies the gaming chip **3** is attached to each of the gaming chips **3**, but as a modification, in place thereof or in addition thereto, the chip RFID system **700** may be constructed by attaching the RFID tag to the chip case **100**. In such a case, chip case ID that uniquely identifies the chip case is stored in the RFID tag attached to each of the chip cases **100**.

(83) Also in these modifications, mutual interference can be prevented by adopting different

frequencies of radio waves used by the card RFID system **600** and the chip RFID system **700**.

(84) Further, in the above embodiments and their modifications, the RFID tag is caused to store code information that uniquely identifies an item (the package **302**, the gaming chip **3** and the like) to which the RFID tag is attached, but information stored in the RFID tag may be other information. For example, the RFID tag may be caused to store information indicating the type of an item to which the RFID tag is attached (for example, information indicating a package to the RFID tag **306** attached to the package **302** and information indicating a gaming chip to the RFID tag **125** contained in the gaming chip **3**). Also in this case, the numbers of the packages **302** and the playing cards **6** can be grasped by the card RFID system **600** and the numbers of the gaming chips **3** and the chip cases **100** can be grasped by the chip RFID system **700**.

(85) Also in the above embodiments, frequencies of the UHF band are used for the card RFID system **600** and frequencies of the HF band are used for the chip RFID system **700**, but as long as frequency bands used for the card RFID system **600** and the chip RFID system **700** are different, frequencies of radio waves used for the card RFID system **600** and the chip RFID system **700** are not limited to the above example.

(86) Also in the above embodiments, the card RFID system **600** adopts the radio wave type and the chip RFID system **700** adopts the electromagnetic induction type, but the types of the card RFID system **600** and the chip RFID system **700** are not limited to the above types and appropriate types may be adopted in accordance with the frequency band of radio waves to be used and other factors.

(87) To solve the above conventional problems, a system that manages packages of shuffled playing cards and gaming chips according to the present invention includes shuffled playing cards having playing cards constituting a predetermined number of decks shuffled in random order and integrally constituted individually as a cage that manages one container or package with a unique playing card ID code attached to the cage managing the container or package, a chip case housing gaming chips having a chip ID code and to which a case ID code is attached, a game table on which a game is played using the shuffled playing cards and the gaming chips, a storage box installed beside the game table to store a plurality of the shuffled playing cards carried from a card room and inserted into a card shooter apparatus on the game table and also to store a plurality of the chip cases housing the gaming chips used on the game table and including an opening/closing mechanism enabling taking out of the shuffled playing cards and the chip tray, and a control apparatus to manage the shuffled playing cards and the gaming chips, the storage box includes one or a plurality of card readers that reads playing card ID codes of all stored shuffled playing cards and also one or a plurality of chip readers that reads case ID codes of all stored chip cases, and the control apparatus has a function to output total numbers of the shuffled playing cards and the chip cases and also all the playing card ID codes and the case ID codes stored in the storage box by monitoring the playing card ID codes read by the card reader and the case ID codes read by the chip reader.

(88) Further, the storage box includes a lock unit configured to prevent taking out of the shuffled playing cards and the chip cases of gaming chips from the storage box.

(89) Further, the storage box may have a shuffled playing card storage box that stores the shuffled playing cards and a chip storage box that houses the gaming chips by allowing the gaming chips to be taken in or out independently.

(90) Further, the case ID code of the chip case is associated with the chip ID code of the gaming chip in the case and the control apparatus has a function to output a total amount of value of all the gaming chips housed in the storage box by acquiring all the case ID codes housed in the storage box.

(91) Further, the control apparatus has a function to grasp an increase/decrease of the chip cases housed in the storage box by periodically monitoring the case ID code of the chip case and, when the increase/decrease is grasped, to output the total amount after the increase/decrease of the value of all the gaming chips housed in the storage box.

(92) To solve the above conventional problems, a system that manages packages of shuffled playing cards and gaming chips according to the present invention may be configured as described below: A system including shuffled playing cards in which playing cards constituting a predetermined number of decks are shuffled in random order and which are integrally constituted individually as one container or package with a unique playing card ID code attached to the container or package, a chip case housing gaming chips having a chip ID code, a game table on which a game is played using the shuffled playing cards and the gaming chips, a storage box installed beside the game table to store a plurality of the shuffled playing cards carried from a card room and inserted into a card shooter apparatus on the game table and also to store a plurality of the chip cases housing the gaming chips used on the game table and including an opening/closing mechanism enabling taking out of the shuffled playing cards and the chip tray, and a control apparatus to manage the shuffled playing cards and the gaming chips, the storage box includes one or a plurality of card readers that reads playing card ID codes of all stored shuffled playing cards and also one or a plurality of chip readers that reads chip ID codes of all stored gaming chips, and the control apparatus has a function to output total numbers of the shuffled playing cards and the gaming chips and also all the playing card ID codes and the chip ID codes stored in the storage box by monitoring the playing card ID codes read by the card reader and the chip ID codes read by the chip reader.

(93) Further, the control apparatus has a function to output a total amount of value of all the gaming chips housed in the storage box by reading all the chip ID codes present in the storage box.

(94) Further, the storage box includes a lock unit configured to prevent taking out of the shuffled playing cards and the chip cases of gaming chips from the storage box.

(95) Further, the storage box has a shuffled playing card storage box that stores the shuffled playing cards and a chip storage box that houses the gaming chips by allowing the gaming chips to be taken in or out independently.

(96) Further, the control apparatus may have a function to grasp an increase/decrease of the gaming chips housed in the storage box by periodically monitoring the chip ID code of the gaming chip stored in the storage box and, when the increase/decrease is grasped, to output the total amount after the increase/decrease of the value of all the gaming chips housed in the storage box.

(97) To solve the above conventional problems, a storage box according to the present invention is a storage box that manages shuffled playing cards and gaming chips, wherein the storage box is carried from a card room to store a plurality of shuffled playing cards and also makes available the shuffled playing cards by individually taking out and inserting the shuffled playing cards into a card shooter apparatus on a game table and further stores a plurality of chip cases housing gaming chips used on the game table to adjust a quantity of the gaming chips on the game table using the chip cases when the gaming chips on the game table are excessive or lacking in accordance with development of a game on the game table and also to be able to store the gaming chips that are excessive before being transferred to a cage that manages the gaming chips of a casino and includes an opening/closing mechanism arranged near the game table to enable taking out of the shuffled playing cards and gaming chips when necessary, the shuffled playing cards have playing cards constituting a predetermined number of decks shuffled in random order and are integrally constituted individually as one container or package with a unique playing card ID code attached to the container or package, the gaming chip has a chip ID code and is housed in the chip case to which a case ID code is attached, the storage box includes one or a plurality of card readers that reads the playing card ID code of all the shuffled playing card stored and also one or a plurality of chip readers that reads the case ID code of all the chip cases stored or the chip ID code of the gaming chips in the chip cases, and a control apparatus has a function to output total numbers of the shuffled playing cards and the chip cases stored in the storage box and also all the playing card ID codes and the case ID codes or chip ID codes stored in the storage box by monitoring the playing card ID codes read by the card reader and the case ID codes or chip ID codes read by the

chip reader.

(98) Further, the storage box may include a lock unit configured to prevent taking out of the shuffled playing cards or the gaming chips from the storage box.

(99) Further, the storage box may include a shuffled playing card storage box that stores the shuffled playing cards and a chip storage box that houses the gaming chips by allowing the gaming chips to be taken in or out independently.

(100) Further, the control apparatus may have a function to output a total amount of value of all the gaming chips housed in the storage box by reading all the chip ID codes present in the storage box.

(101) Further, the case ID code of the chip case may be associated with the chip ID code of the gaming chip in the case and the control apparatus may have a function to output a total amount of value of all the gaming chips housed in the storage box based on the case ID code by acquiring all the case ID codes housed in the storage box.

(102) To solve the above conventional problems, a system that manages gaming chips according to the present invention includes a chip case that houses gaming chips having a chip ID code, a storage box that stores a plurality of the chip cases housing the gaming chips used on a game table, adjusts a quantity of the gaming chips on a chip float of the game table using the chip cases when the gaming chips placed on the chip float of the game table are excessive or lacking in accordance with development of a game on the game table and also is able to store the gaming chips that are excessive before being transferred to a cage that manages the gaming chips of a casino and includes an opening/closing mechanism arranged beside the game table to enable taking out of shuffled playing cards and the chip tray when necessary, the game table on which a game is played using the gaming chips, and a control apparatus to manage the gaming chips, wherein the storage box includes one or a plurality of chip readers that reads chip ID codes of all stored gaming chips, and the control apparatus has a function to output a total number of the gaming chips stored in the storage box and all the chip ID codes stored in the storage box by monitoring the chip ID code read by the chip reader.

(103) Further, the control apparatus may have a function to output a total amount of value of all the gaming chips housed in the storage box by reading all the chip ID codes present in the storage box.

(104) Further, the storage box may include a lock unit configured to prevent taking out of the chip cases of gaming chips from the storage box.

(105) Further, the control apparatus may have a function to grasp an increase/decrease of the gaming chips housed in the storage box by periodically monitoring the chip ID code of the gaming chip stored in the storage box and, when the increase/decrease is grasped, to output the total amount after the increase/decrease of the value of all the gaming chips housed in the storage box.

(106) Further, the control apparatus may have a function to output whether an increase/decrease amount or increase/decrease value of the gaming chips placed on the chip float of the game table or an increase/decrease amount or increase/decrease value of the gaming chips placed in the cage that manages the gaming chips of a casino and an increase/decrease amount or increase/decrease value of the gaming chips housed in the storage box match.

(107) Further, a case ID code may be attached to the chip case and associated with the chip ID code of the gaming chip in the case.

Claims

1. A system configured to manage gaming chips and packages of shuffled playing cards, the system comprising: a set of shuffled playing cards, wherein the set of playing cards constituting a predetermined number of decks are shuffled in a random order, wherein a unique card ID is assigned to the set of shuffled playing cards using a wireless tag; a gaming chip having a chip ID using a wireless tag; a game table where games are played using the shuffled playing cards and the gaming chips; a card storage area where a plurality of shuffled playing cards carried from a card

room and to be loaded to a card shooter device on the game table are stored; a chip storage area where a plurality of gaming chips to be used on the game table are stored; one or more card readers configured to read the card IDs of all the set of shuffled playing cards stored in the card storage area; and one or more chip readers configured to read the chip IDs of all the chips stored in the chip storage area, wherein an information transmission method of the one or more card readers and an information transmission method of the one or more chip readers are different from each other, so that the one or more card readers can read the wireless tag for the card ID and cannot read the wireless tag for the chip ID, and the one or more chip readers can read the wireless tag for the chip ID and cannot read the wireless tag for the card ID.

2. The system according to claim 1, wherein the card ID is stored in a UHF wireless tag and the chip ID is stored in a HF wireless tag.

3. The system according to claim 1, further comprising a controller connected to the one or more card readers and the one or more chip readers, and configured to detect any variation in the information read by the one or more card readers or the one or more chip readers.

4. The system according to claim 1, wherein the one or more card readers are configured to read information stored in the wireless tag using a radio wave method, and the one or more chip readers are configured to read information stored in the wireless tag using an electromagnetic induction method.

5. A system configured to manage playing cards to which card information is attached by RFID tag for card and gaming chips to which chip information is attached by RFID tag for chip, the system comprising: a card storage configured to store the playing cards; a chip storage configured to store the gaming chips; an RFID reader for cards configured to read information stored in the RFID tag for card attached to the playing card stored in the card storage; and an RFID reader for chips configured to read information stored in the RFID tag for chip attached to the gaming chip stored in the chip storage, wherein the RFID reader for cards can read the RFID tag for card and cannot read the RFID tag for chip, and the RFID reader for chips can read the RFID tag for chip and cannot read the RFID tag for card.

6. The system according to claim 5, wherein the card information is assigned to a package comprising a plurality of the playing cards, and the chip information is assigned to each of the gaming chips.

7. The system according to claim 6, wherein the RFID tag for card is attached to the package, and the RFID tag for chip is embedded in the gaming chip.

8. The system according to claim 5, wherein the RFID tag for card is embedded in the playing card, and the card information is assigned to each of the playing cards.

9. The system according to claim 5, further comprising a controller connected to the RFID reader for cards and the RFID reader for chips, and detecting any variation in the information read by the RFID reader for cards or the RFID reader for chips.

10. The system according to claim 5, wherein the RFID reader for cards is configured to read information on the RFID tag for card using a radio wave method, and the RFID reader for chips is configured to read information on the RFID tag for chip using an electromagnetic induction method.
